

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{dy}{dx} = \frac{(x+1)^2 \times (10x+10) - (5x^2 + 10x) \times 2(x+1)}{(x+1)^4}$	M1	This mark is given for an attempt to differentiate the expression for y
		A1	This mark is given for correctly differentiating the expression for y
	$\frac{dy}{dx} = \frac{(x+1) \times (10x+10) - (5x^2 + 10x) \times 2}{(x+1)^3}$	M1	This mark is given for cancelling the expression through by $(x+1)$
	$\frac{dy}{dx} = \frac{10}{(x+1)^3}$	A1	This mark is given for a fully correct expression for $\frac{dy}{dx}$
(b)	If $A > 0$ and $n = 1, 3$ then $x < -1$	B1	This mark is given for deducing that $\frac{dy}{dx} < 0 \Rightarrow x < -1$.

Q2.

Question	Scheme	Marks	AOs
(a)	$x = -4$ or $y = -5$	B1	1.1b
(b)	$P(-4, -5)$	B1	2.2a
		(2)	
	$3x + 40 = -2(x + 4) - 5 \Rightarrow x = \dots$	M1	1.1b
	$x = -10.6$	A1	2.1
(c)	$a > 2$	B1	2.2a
	$y = ax \Rightarrow -5 = -4a \Rightarrow a = \frac{5}{4}$	M1	3.1a
	$\{a : a \leq 1.25\} \cup \{a : a > 2\}$	A1	2.5
		(3)	
(7 marks)			

Notes:

- (a)
B1: One correct coordinate. Either $x = -4$ or $y = -5$ or $(-4, \dots)$ or $(\dots, -5)$ seen.
B1: Deduces that $P(-4, -5)$ Accept written separately e.g. $x = -4$, $y = -5$
- (b)
M1: Attempts to solve $3x + 40 = -2(x + 4) - 5 \Rightarrow x = \dots$ Must reach a value for x .
 You may see the attempt crossed out but you can still take this as an attempt to solve the required equation.
A1: $x = -10.6$ or e.g. $-\frac{53}{5}$ only. If other values are given, e.g. $x = -37$ they must be rejected or the $-\frac{53}{5}$ clearly chosen as their answer. Ignore any attempts to find y .
 Alternative by squaring:

$$3x + 40 = 2|x + 4| - 5 \Rightarrow 3x + 45 = 2|x + 4| \Rightarrow 9x^2 + 270x + 2025 = 4(x^2 + 8x + 16)$$

$$\Rightarrow 5x^2 + 238x + 1961 = 0 \Rightarrow x = -37, -\frac{53}{5}$$
M1 for isolating the $|x + 4|$, squaring both sides and solving the resulting quadratic
A1 for selecting the $-\frac{53}{5}$
- Correct answer with no working scores both marks.
- (c)
B1: Deduces that $a > 2$
M1: Attempts to find a value for a using their $P(-4, -5)$
 Alternatively attempts to solve $ax = 2(x + 4) - 5$ and $ax = 2(x + 4) - 5$ to obtain a value for a .
A1: Correct range in acceptable set notation.
 $\{a : a \leq 1.25\} \cup \{a : a > 2\}$
 $\{a : a \leq 1.25\}, \{a : a > 2\}$
 Examples: $\{a : a \leq 1.25 \text{ or } a > 2\}$
 $\{a : a \leq 1.25, a > 2\}$
 $(-\infty, 1.25] \cup (2, \infty)$
 $(-\infty, 1.25], (2, \infty)$

Q3.

Question	Scheme	Marks	AOs
	Sets $f(-2) = 0 \Rightarrow (-2 - 4)((-2)^2 - 3 \times -2 + k) - 42 = 0$	M1	3.1a
	$-6(k + 10) = 42 \Rightarrow k = \dots$	M1	1.1b
	$k = -17$	A1	1.1b
		(3)	
(3 marks)			
Notes:			

M1: Attempts $f(-2) = 0$ leading to an equation in k . So $(-2-4)((-2)^2 - 3 \times -2 + k) - 42 = 0$ is fine

Condone slips but expect to see a first bracket of $(-2-4)$.

"- 42" must not be omitted but could appear as +42 with a sign slip.

There may have been attempts to expand $f(x) = (x-4)(x^2 - 3x + k) - 42$ before attempting to set $f(-2) = 0$. This is acceptable and condone slips/errors in the expansion, but the 42 must be present. FYI the expanded (and simplified) $f(x) = x^3 - 7x^2 + (12+k)x - 4k - 42$

M1: Solves a linear equation in k as a result of setting $f(\pm 2) = 0$.

The ± 42 must be there at some point when the substitution is made.

Allow minimal evidence here. A linear equation leading to a solution is fine.

If $f(x)$ is expanded then it is dependent upon being a cubic which contains a kx term and a '42'

A1: $k = -17$ correct answer following correct work but allow recovery from invisible brackets

Answers of $k = -17$ may appear with very little or no working, perhaps via trial and improvement. If so, then marks can only be allocated if evidence is shown.

E.g. $k = -17 \Rightarrow f(x) = (x-4)(x^2 - 3x - 17) - 42$

$f(-2) = (-6) \times (-7) - 42 = 0$. Hence $(x+2)$ is a factor.

More difficult alternative methods may be seen

Alt I: You may see attempts via division / inspection

$$\begin{array}{r}
 x^2 - 9x + (k+30) \\
 x+2 \overline{) x^3 - 7x^2 + (12+k)x - 4k - 42} \\
 \underline{x^2 + 2x} \\
 -9x + (k+30) \\
 \underline{-9x - 18} \\
 -6k - 102
 \end{array}$$

Then sets remainder $-6k - 102 = 0 \Rightarrow k = -17$

M1: For dividing their cubic by $(x+2)$ which has both an x and a constant coefficient in k , leading to a quadratic quotient and a linear remainder in k which is then set $= 0$

M1: Solves a equation resulting from setting a linear remainder in k equal to 0. It is dependent on the first M via this route

A1: Completely correct with $k = -17$

Alt II: You may also see a grid or an attempt at factorisation via inspection

$$\begin{array}{r|rrrr}
 & x^2 & -9x & -2k-21 & \\
 x & x^3 & -9x^2 & (-2k-21)x & \\
 +2 & 2x^2 & -18x & -4k-42 &
 \end{array}$$

OR $x^3 - 7x^2 + (12+k)x - 4k - 42 \equiv (x+2)(x^2 - 9x - 2k - 21)$

which should be followed by equating the x terms to form an equation in k

$$12 + k = -18 - 2k - 21 \Rightarrow 3k = -51 \Rightarrow k = -17$$

OR $x^3 - 7x^2 + (12+k)x - 4k - 42 \equiv (x+2)(x^2 - 9x + k + 30)$

which should be followed by equating the constant terms to form an equation in k

$$-4k - 42 = 2(k + 30) \Rightarrow 6k = -102 \Rightarrow k = -17$$

The above are examples. There may be other correct attempts so look at what is done.

M1: For an attempt at factorising E.g. $x^3 - 7x^2 + (12+k)x - 4k - 42 \equiv (x+2)(x^2 + bx + c)$ and attempting to set up three equations in b , c and k . E.g. $2 + b = -7$, $2b + c = 12 + k$, $2c = -4k - 42$

The expanded $f(x)$ must be a cubic which contains both a kx term and a '42'

M1: Solves the equations set up from an allowable equation to find k . It is dependent via this route.

A1: Completely correct with $k = -17$

Q4.

Part	Working or answer an examiner might expect to see	Mark	Notes
	$f(-3) = 3 \times (-3)^2 + 2a \times (-3)^2 - 4 \times -3 + 5a = 0$	M1	This mark is given for a method to set $f(-3) = 0$
	$f(-3) = 23a - 69 = 0$ $23a = 69$	M1	This mark is given for finding an equation to solve for a
	$a = 3$	A1	This mark is given for finding the correct value of a

Q5.

Question	Scheme	Marks	AOs
	$f(1) = a(1)^3 + 10(1)^2 - 3a(1) - 4 = 0$	M1	3.1a
	$6 - 2a = 0 \Rightarrow a = \dots$	M1	1.1b
	$a = 3$	A1	1.1b
		(3)	
(3 marks)			
Notes			

Main method seen:

M1: Attempts $f(1) = 0$ to set up an equation in a . It is implied by $a + 10 - 3a - 4 = 0$

Condone a slip but attempting $f(-1) = 0$ is M0

M1: Solves a linear equation in a .

Using the main method it is dependent upon having set $f(\pm 1) = 0$

It is implied by a solution of $\pm a \pm 10 \pm 3a \pm 4 = 0$.

Don't be concerned about the mechanics of the solution.

A1: $a = 3$ (following correct work)

Answers without working scores 0 marks. The method must be made clear. Candidates cannot guess.

However if a candidate states for example, when $a = 3$, $f(x) = 3x^3 + 10x^2 - 9x - 4$ and shows that $(x-1)$ is a factor of this $f(x)$ by an allowable method, they should be awarded M1 M1 A1

E.g. 1: $3x^3 + 10x^2 - 9x - 4 = (x-1)(3x^2 + 13x + 4)$ Hence $a = 3$

E.g. 2: $f(x) = 3x^3 + 10x^2 - 9x - 4$, $f(1) = 3 + 10 - 9 - 4 = 0$ Hence $a = 3$

The solutions via this method must end with the value for a to score the A1

Other methods are available. They are more difficult to determine what the candidate is doing.

Please send to review if you are uncertain

It is important that a correct method is attempted so look at how the two M's are scored

Amongst others are:

Alt (1) by inspection which may be seen in a table

	ax^2	$(10+a)x$	4
x	ax^3	$(10+a)x^2$	$4x$
-1	$-ax^2$	$-(10+a)x$	-4

$$ax^3 + 10x^2 - 3ax - 4 = (x-1)(ax^2 + (10+a)x + 4) \quad \text{and sets terms in } x \text{ equal}$$

$$-3a = -(10+a) + 4 \Rightarrow 2a = 6 \Rightarrow a = 3$$

M1: This method is implied by a correct equation, usually $-3a = -(10+a) + 4$

M1: Attempts to find the quadratic factor which must be of the form $ax^2 + g(a)x \pm 4$ and then forms and solves a linear equation formed by linking the coefficients or terms in x

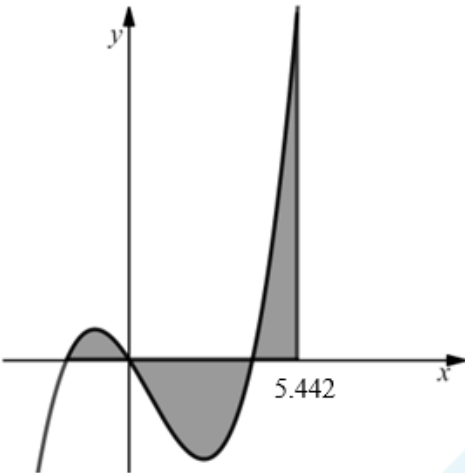
Alt (2) By division:
$$\begin{array}{r}
 ax^2 + (\pm 10 \pm a)x + (10 - 2a) \\
 x-1 \overline{) ax^3 + 10x^2 - 3ax - 4} \\
 \underline{ax^3 - ax^2} \\
 (10+a)x^2 - 3ax \\
 \underline{(10+a)x^2 - (10+a)x} \\
 (-2a+10)x
 \end{array}$$

M1: This method is implied by a correct equation, usually $-10 + 2a = -4$

M1: Attempts to divide with quotient of $ax^2 + (\pm 10 \pm a)x + h(a)$ and then forms and solves a linear equation in a formed by setting the remainder = 0.

Q6.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$y = x(x+2)(x-4) = x^3 - 2x^2 - 8x$	B1	This mark is given for expanding brackets as a first step to a solution
	$\int_{-2}^0 x^3 - 2x^2 - 8x \, dx$	M1	This mark is given for a method to find the exact area of R_1
	$= \left[\frac{1}{4}x^4 - \frac{2}{3}x^3 - 4x^2 \right]_{-2}^0$	M1	This mark is given for a method to evaluate the integral
	$= 0 - (4 - \frac{16}{3} - 16) = \frac{20}{3}$	A1	This mark is given for a full method to show the exact value of R_1
(b)	$\frac{1}{4}b^4 - \frac{2}{3}b^3 - 4b^2 = -\frac{20}{3}$	M1	This mark is given for deducing the area of $R_2 = -\frac{20}{3}$
	$3b^4 - 8b^3 - 48b^2 + 80 = 0$	A1	This mark is given for rearranging the equation to a quartic
	$(b+2)^2(3b^2 - 20b + 20)$ $= (b^2 + 4b + 4)(3b^2 - 20b + 20)$	M1	This mark is given for expanding the equation given
	$= 3b^4 - 8b^3 - 48b^2 + 80 = 0$ The two equations are the same, so verified	A1	This mark is for showing, and stating, that the equations are the same

(c)		B1	This mark is given for a sketch of the curve with $b = 5.442$ shown
	Between $x = -2$ and $b = 5.442$, the area above the x -axis is the same as the area below the x -axis	B1	This mark is given for a valid explanation of the significance of the root 5.442

Q7.

Question	Scheme	Marks	AOs
	$\frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv A + \frac{B}{(x-3)} + \frac{C}{(1-2x)}$		
(a) Way 1	$1+11x-6x^2 \equiv A(1-2x)(x-3) + B(1-2x) + C(x-3) \Rightarrow B = \dots, C = \dots$	M1	2.1
	$A = 3$	B1	1.1b
	Uses substitution or compares terms to find either $B = \dots$ or $C = \dots$	M1	1.1b
	$B = 4$ and $C = -2$ which have been found using a correct identity	A1	1.1b
		(4)	
(a) Way 2	{long division gives} $\frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv 3 + \frac{-10x+10}{(x-3)(1-2x)}$		
	$-10x+10 \equiv B(1-2x) + C(x-3) \Rightarrow B = \dots, C = \dots$	M1	2.1
	$A = 3$	B1	1.1b
	Uses substitution or compares terms to find either $B = \dots$ or $C = \dots$	M1	1.1b
	$B = 4$ and $C = -2$ which have been found using $-10x+10 \equiv B(1-2x) + C(x-3)$	A1	1.1b
		(4)	
(b)	$f(x) = 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)}$ { $= 3 + 4(x-3)^{-1} - 2(1-2x)^{-1}$ } ; $x > 3$		
	$f'(x) = -4(x-3)^{-2} - 4(1-2x)^{-2} \left\{ = -\frac{4}{(x-3)^2} - \frac{4}{(1-2x)^2} \right\}$	M1 A1ft	2.1 1.1b
	Correct $f'(x)$ and as $(x-3)^2 > 0$ and $(1-2x)^2 > 0$, then $f'(x) = -(+ve) - (+ve) < 0$, so $f(x)$ is a decreasing function	A1	2.4
		(3)	
(7 marks)			

Notes for Question	
(a)	
M1:	Way 1: Uses a correct identity $1+11x-6x^2 \equiv A(1-2x)(x-3) + B(1-2x) + C(x-3)$ in a complete method to find values for B and C . Note: Allow one slip in copying $1+11x-6x^2$ Way 2: Uses a correct identity $-10x+10 \equiv B(1-2x) + C(x-3)$ (which has been found from long division) in a complete method to find values for B and C
B1:	$A = 3$
M1:	Attempts to find the value of either B or C from their identity This can be achieved by <i>either</i> substituting values into their identity <i>or</i> by comparing coefficients and solving the resulting equations simultaneously
A1:	See scheme
Note:	Way 1: Comparing terms: $x^2: -6 = -2A; x: 11 = 7A - 2B + C; \text{constant: } 1 = -3A + B - 3C$ Way 1: Substituting: $x = 3: -20 = -5B \Rightarrow B = 4; x = \frac{1}{2}: 5 = -\frac{5}{2}C \Rightarrow C = -2$
Note:	Way 2: Comparing terms: $x: -10 = -2B + C; \text{constant: } 10 = B - 3C$ Way 2: Substituting: $x = 3: -20 = -5B \Rightarrow B = 4; x = \frac{1}{2}: 5 = -\frac{5}{2}C \Rightarrow C = -2$
Note:	$A = 3, B = 4, C = -2$ from no working scores M1B1M1A1
Note:	The final A1 mark is effectively dependent upon both M marks

Notes for Question Continued	
(a) ctd	
Note:	Writing $1+11x-6x^2 \equiv B(1-2x) + C(x-3) \Rightarrow B = 4, C = -2$ will get 1 st M0, 2 nd M1, 1 st A0
Note:	Way 1: You can imply a correct identity $1+11x-6x^2 \equiv A(1-2x)(x-3) + B(1-2x) + C(x-3)$ from seeing $\frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv \frac{A(1-2x)(x-3) + B(1-2x) + C(x-3)}{(x-3)(1-2x)}$
Note:	Way 2: You can imply a correct identity $-10x+10 \equiv B(1-2x) + C(x-3)$ from seeing $\frac{-10x+10}{(x-3)(1-2x)} \equiv \frac{B(1-2x) + C(x-3)}{(x-3)(1-2x)}$
(b)	
M1:	Differentiates to give $\{f'(x) = \} \pm \lambda(x-3)^{-2} \pm \mu(1-2x)^{-2}; \lambda, \mu \neq 0$
A1ft:	$f'(x) = -4(x-3)^{-2} - 4(1-2x)^{-2}$, which can be simplified or un-simplified
Note:	Allow A1ft for $f'(x) = -(\text{their } B)(x-3)^{-2} + (2)(\text{their } C)(1-2x)^{-2}; (\text{their } B), (\text{their } C) \neq 0$
A1:	$f'(x) = -4(x-3)^{-2} - 4(1-2x)^{-2}$ or $f'(x) = -\frac{4}{(x-3)^2} - \frac{4}{(1-2x)^2}$ and a correct explanation e.g. $f'(x) = -(+ve) - (+ve) < 0$, so $f(x)$ is a decreasing {function}
Note:	The final A mark can be scored in part (b) from an incorrect $A = \dots$ or from $A = 0$ or no value of A found in part (a)

Notes for Question Continued - Alternatives			
(a)			
Note:	Be aware of the following alternative solutions, by initially dividing by " $(x-3)$ " or " $(1-2x)$ "		
	$\bullet \frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv \frac{-6x-7}{(1-2x)} - \frac{20}{(x-3)(1-2x)} \equiv 3 - \frac{10}{(1-2x)} - \frac{20}{(x-3)(1-2x)}$ $\frac{20}{(x-3)(1-2x)} \equiv \frac{D}{(x-3)} + \frac{E}{(1-2x)} \Rightarrow 20 \equiv D(1-2x) + E(x-3) \Rightarrow D = -4, E = -8$ $\Rightarrow 3 - \frac{10}{(1-2x)} - \left(\frac{-4}{(x-3)} + \frac{-8}{(1-2x)} \right) \equiv 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)}; A=3, B=4, C=-2$ $\bullet \frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv \frac{3x-4}{(x-3)} + \frac{5}{(x-3)(1-2x)} \equiv 3 + \frac{5}{(x-3)} + \frac{5}{(x-3)(1-2x)}$ $\frac{5}{(x-3)(1-2x)} \equiv \frac{D}{(x-3)} + \frac{E}{(1-2x)} \Rightarrow 5 \equiv D(1-2x) + E(x-3) \Rightarrow D = -1, E = -2$ $\Rightarrow 3 + \frac{5}{(x-3)} + \left(\frac{-1}{(x-3)} + \frac{-2}{(1-2x)} \right) \equiv 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)}; A=3, B=4, C=-2$		
(b)			
	Alternative Method 1:		
	$f(x) = \frac{1+11x-6x^2}{(x-3)(1-2x)}, x > 3 \Rightarrow f(x) = \frac{1+11x-6x^2}{-2x^2+7x-3}; \begin{cases} u = 1+11x-6x^2 & v = -2x^2+7x-3 \\ u' = 11-12x & v' = -4x+7 \end{cases}$		
	$f'(x) = \frac{(-2x^2+7x-3)(11-12x) - (1+11x-6x^2)(-4x+7)}{(-2x^2+7x-3)^2}$	Uses quotient rule to find $f'(x)$	M1
		Correct differentiation	A1
	$f'(x) = \frac{-20((x-1)^2+1)}{(-2x^2+7x-3)^2}$ and a correct explanation,		A1
	e.g. $f'(x) = -\frac{(+ve)}{(+ve)} < 0$, so $f(x)$ is a decreasing {function}		
	Alternative Method 2:		
	Allow M1A1A1 for the following solution:		
	Given $f(x) = 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)} = 3 + \frac{4}{(x-3)} + \frac{2}{(2x-1)}$		
	as $\frac{4}{(x-3)}$ decreases when $x > 3$ and $\frac{2}{(2x-1)}$ decreases when $x > 3$		
	then $f(x)$ is a decreasing {function}		

Q8.

Question	Scheme	Marks	AOs
(a)	$k = e^2$ or $x \neq e^2$	B1	2.2a
		(1)	

(b)	$g'(x) = \frac{(\ln x - 2) \times \frac{3}{x} - (3 \ln x - 7) \times \frac{1}{x}}{(\ln x - 2)^2} = \frac{1}{x(\ln x - 2)^2}$ or $g'(x) = \frac{d}{dx} \left(3 - (\ln(x) - 2)^{-1} \right) = (\ln x - 2)^{-2} \times \frac{1}{x} = \frac{1}{x(\ln x - 2)^2}$ or $g'(x) = (\ln x - 2)^{-1} \times \frac{3}{x} - (3 \ln x - 7)(\ln x - 2)^{-2} \times \frac{1}{x} = \frac{1}{x(\ln x - 2)^2}$	M1 A1	1.1b 2.1
	As $x > 0$ (or $1/x > 0$) AND $\ln x - 2$ is squared so $g'(x) > 0$	A1cso	2.4
		(3)	
(c)	Attempts to solve either $3 \ln x - 7 \dots 0$ or $\ln x - 2 \dots 0$ or $3 \ln a - 7 \dots 0$ or $\ln a - 2 \dots 0$ where $\dots$ is " $=$ " or " $>$ " to reach a value for x or a but may be seen as an inequality e.g. $x > \dots$ or $a > \dots$	M1	3.1a
	$0 < a < e^2, \quad a > e^{\frac{7}{3}}$	A1	2.2a
		(2)	
(6 marks)			

Notes:

(a)

B1: Deduces $k = e^2$ or $x \neq e^2$ Condone $k = \text{awrt } 7.39$ or $x \neq \text{awrt } 7.39$

(b)

M1: Attempts to differentiate via the quotient rule and with $\ln x \rightarrow \frac{1}{x}$ so allow for:

$$\frac{d}{dx}(g(x)) = \frac{(\ln x - 2) \times \frac{\alpha}{x} - (3 \ln x - 7) \times \frac{\beta}{x}}{(\ln x - 2)^2}, \quad \beta > 0$$

But a correct rule may be implied by their u, v, u', v' followed by applying $\frac{vu' - uv'}{v^2}$ etc.

Alternatively attempts to write $g(x) = \frac{3 \ln(x) - 7}{\ln(x) - 2} = 3 - (\ln(x) - 2)^{-1}$ and attempts the chain rule so allow for:

$$3 - (\ln(x) - 2)^{-1} \rightarrow (\ln(x) - 2)^{-2} \times \frac{\alpha}{x}$$

Alternatively writes $g(x) = (3 \ln(x) - 7)(\ln(x) - 2)^{-1}$ and attempts the product rule so allow for:

$$g'(x) = (\ln x - 2)^{-1} \times \frac{\alpha}{x} - (3 \ln x - 7)(\ln x - 2)^{-2} \times \frac{\beta}{x}$$

In general condone missing brackets for the M mark. E.g. if they quote $u = 3 \ln x - 7$ and $v = \ln x - 2$ and don't make the differentiation easier, they can be awarded this mark for applying the correct rule.

Also allow this mark if they quote the correct quotient rule but only have v rather than v^2 in the denominator.

A1: $\frac{1}{x(\ln x - 2)^2}$ Allow $\frac{\frac{1}{x}}{(\ln x - 2)^2}$ i.e. we need to see the numerator simplified to $1/x$

Note that some candidates establish the correct numerator and correct denominator independently and provided they obtain the correct expressions, this mark can be awarded.

But allow a correctly expanded denominator.

A1cso: States that as $x > 0$ AND $\ln x - 2$ is squared so $g'(x) > 0$

(c)

M1: Attempts to solve either $3 \ln x - 7 = 0$ or $\ln x - 2 = 0$ or using inequalities e.g. $3 \ln x - 7 > 0$

A1: $0 < a < e^2, \quad a > e^{\frac{7}{3}}$

Q9.

Question	Scheme	Marks	AOs
	Attempts equation of line Eg Substitutes $(-2, 13)$ into $y = mx + 25$ and finds m	M1	1.1b
	Equation of l is $y = 6x + 25$	A1	1.1b
	Attempts equation of C Eg Attempts to use the intercept $(0, 25)$ within the equation $y = a(x \pm 2)^2 + 13$, in order to find a	M1	3.1a
	Equation of C is $y = 3(x+2)^2 + 13$ or $y = 3x^2 + 12x + 25$	A1	1.1b
	Region R is defined by $3(x+2)^2 + 13 < y < 6x + 25$ o.e.	B1ft	2.5
		(5)	
(5 marks)			

The first two marks are awarded for finding the equation of the line

M1: Uses the information in an attempt to find an equation for the line l .

E.g. Attempt using two points: Finds $m = \pm \frac{25-13}{2}$ and uses of one of the points in their $y = mx + c$ or equivalent to find c . Alternatively uses the intercept as shown in main scheme.

A1: $y = 6x + 25$ seen or implied. This alone scores the first two marks. Do not accept $l = 6x + 25$

It must be in the form $y = \dots$ but the correct equation can be implied from an inequality. E.g. $\dots < y < 6x + 25$

The next two marks are awarded for finding the equation of the curve

M1: A complete method to find the constant a in $y = a(x \pm 2)^2 + 13$ or the constants a, b in $y = ax^2 + bx + 25$.

An alternative to the main scheme is deducing equation is of the form $y = ax^2 + bx + 25$ and setting and solving a pair of simultaneous equations in a and b using the point $(-2, 13)$ the gradient being 0 at $x = -2$. Condone slips. Implied by $C = 3x^2 + 12x + 25$ or $3x^2 + 12x + 25$

FYI the correct equations are $13 = 4a - 2b + 25$ ($2a - b = -6$) and $-4a + b = 0$

A1: $y = 3(x+2)^2 + 13$ or equivalent such as $y = 3x^2 + 12x + 25$, $f(x) = 3(x+2)^2 + 13$.

Do not accept $C = 3x^2 + 12x + 25$ or just $3x^2 + 12x + 25$ for the A1 but may be implied from an inequality or from an attempt at the area, E.g. $\int 3x^2 + 12x + 25 \, dx$

B1ft: Fully defines the region R . Follow through on their equations for l and C .

Allow strict or non -strict inequalities as long as they are used consistently.

E.g. Allow for example " $3(x+2)^2 + 13 < y < 6x + 25$ " " $3(x+2)^2 + 13 \leq y \leq 6x + 25$ "

Allow the inequalities to be given separately, e.g. $y < 6x + 25, y > 3(x+2)^2 + 13$. Set notation may be used so

$\{(x, y) : y > 3(x+2)^2 + 13\} \cap \{(x, y) : y < 6x + 25\}$ is fine but condone with or without any of $(x, y) \leftrightarrow y \leftrightarrow x$

Incorrect examples include " $y < 6x + 25$ or $y > 3(x+2)^2 + 13$ ", $\{(x, y) : y > 3(x+2)^2 + 13\} \cup \{(x, y) : y < 6x + 25\}$

Values of x could be included but they must be correct. So $3(x+2)^2 + 13 < y < 6x + 25, x < 0$ is fine

If there are multiple solutions mark the final one.

Q10.

Question	Scheme	Marks	AOs
	Attempts to substitute $= \frac{x+1}{2}$ into $y \Rightarrow y = 4\left(\frac{x+1}{2}\right) - 7 + \frac{6}{(x+1)}$	M1	2.1
	Attempts to write as a single fraction $y = \frac{(2x-5)(x+1)+6}{(x+1)}$	M1	2.1
	$y = \frac{2x^2-3x+1}{x+1} \quad a=-3, b=1$	A1	1.1b
(3 marks)			
Notes:			
M1: Score for an attempt at substituting $t = \frac{x+1}{2}$ or equivalent into $y = 4t - 7 + \frac{3}{t}$ M1: Award this for an attempt at a single fraction with a correct common denominator. Their $4\left(\frac{x+1}{2}\right) - 7$ term may be simplified first A1: Correct answer only $y = \frac{2x^2-3x+1}{x+1} \quad a=-3, b=1$			

Q11.

Question	Scheme	Marks	AOs
	Sets $f(-2) = 0 \Rightarrow 2 \times (-2)^3 - 5 \times (-2)^2 + a \times -2 + a = 0$	M1	3.1a
	Solves linear equation $2a - a = -36 \Rightarrow a =$	dM1	1.1b
	$\Rightarrow a = -36$	A1	1.1b
(3 marks)			
Notes:			
M1: Selects a suitable method given that $(x+2)$ is a factor of $f(x)$ Accept either setting $f(-2) = 0$ or attempted division of $f(x)$ by $(x+2)$ dM1: Solves linear equation in a . Minimum requirement is that there are two terms in ' a ' which must be collected to get $..a = .. \Rightarrow a =$ A1: $a = -36$			

Q12.

Question	Scheme	Marks	AOs
	(a) $f(x) = -3x^3 + 8x^2 - 9x + 10, x \in \mathbb{R}$		
(a)	(i) $\{f(2) = -24 + 32 - 18 + 10 \Rightarrow\} f(2) = 0$	B1	1.1b
	(ii) $\{f(x) = (x-2)(-3x^2 + 2x - 5) \text{ or } (2-x)(3x^2 - 2x + 5)\}$	M1	2.2a
		A1	1.1b
		(3)	
(b)	$-3y^6 + 8y^4 - 9y^2 + 10 = 0 \Rightarrow (y^2 - 2)(-3y^4 + 2y^2 - 5) = 0$		
	Gives a partial explanation by - explaining that $-3y^4 + 2y^2 - 5 = 0$ has no {real} solutions with a reason, e.g. $b^2 - 4ac = (2)^2 - 4(-3)(-5) = -56 < 0$ - or stating that $y^2 = 2$ has 2 {real} solutions or $y = \pm\sqrt{2}$ {only}	M1	2.4
	Complete proof that the given equation has exactly two {real} solutions	A1	2.1
		(2)	
(c)	$3 \tan^3 \theta - 8 \tan^2 \theta + 9 \tan \theta - 10 = 0; 7\pi \leq \theta < 10\pi$		
	{Deduces that} there are 3 solutions	B1	2.2a
		(1)	
(6 marks)			

Notes for Question	
(a)(i)	
B1:	$f(2) = 0$ or 0 stated by itself in part (a)(i)
(a)(ii)	
M1:	Deduces that $(x-2)$ or $(2-x)$ is a factor and attempts to find the other quadratic factor by - using long division to obtain either $\pm 3x^2 \pm kx + \dots, k = \text{value} \neq 0$ or $\pm 3x^2 \pm \alpha x + \beta, \beta = \text{value} \neq 0, \alpha$ can be 0 - factorising to obtain their quadratic factor in the form $(\pm 3x^2 \pm kx \pm c), k = \text{value} \neq 0, c$ can be 0, or in the form $(\pm 3x^2 \pm \alpha x \pm \beta), \beta = \text{value} \neq 0, \alpha$ can be 0
A1:	$(x-2)(-3x^2 + 2x - 5), (2-x)(3x^2 - 2x + 5)$ or $-(x-2)(3x^2 - 2x + 5)$ stated together as a product
(b)	
M1:	See scheme
A1:	See scheme. Proof must be correct <i>with no errors</i> , e.g. giving an incorrect discriminant value
Note:	Correct calculation e.g. $(2)^2 - 4(-3)(-5), 4 - 60$ or -56 must be given for the first explanation
Note:	Note that M1 can be allowed for - a correct follow through calculation for the discriminant of their "$-3y^4 + 2y^2 - 5$" which would lead to a value < 0 together with an explanation that $-3y^4 + 2y^2 - 5 = 0$ has no {real} solutions - or for the omission of < 0
Note:	< 0 must also been stated in a discriminant method for A1
Note:	Do not allow A1 for incorrect working, e.g. $(2)^2 - 4(-3)(-5) = -54 < 0$
Note:	$y^2 = 2 \Rightarrow y = \pm 2$, so 2 solutions is not allowed for A1, but can be condoned for M1
Note:	Using the formula on $-3y^4 + 2y^2 - 5 = 0$ or $-3x^2 + 2x - 5 = 0$ gives y^2 or $x = \frac{-2 \pm \sqrt{-56}}{-6}$ or $\frac{-1 \pm \sqrt{-14}}{-3}$
Note:	Completing the square on $-3x^2 + 2x - 5 = 0$ gives $x^2 - \frac{2}{3}x + \frac{5}{3} = 0 \Rightarrow \left(x - \frac{1}{3}\right)^2 - \frac{1}{9} + \frac{5}{3} = 0 \Rightarrow x = \frac{1}{3} \pm \sqrt{\frac{-14}{9}}$
Note:	Do not recover work for part (b) in part (c)
(c)	
B1:	See scheme
Note:	Give B0 for stating $\theta = \text{awrt } 23.1, \text{awrt } 26.2, \text{awrt } 29.4$ without reference to 3 solutions